

Exam :
Subject :
Total Marks :
QP :

End Term Quiz
LLM
50.00
2025 Apr13: IIT M FN EXAM QDB2

Exam Mode

Learning Mode

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QUESTION MENU

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Question 1 : 6406531245585

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Discussions (0)

Suggest Correction

Broken Format?

Question 2 : 6406531245586

Total Mark : 0.00 | Type : MCQ



General Note:

- If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.
- In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

OPTIONS :

- ☐ Instructions has been mentioned above.
- ☐ This Instructions is just for a reference & not for an evaluation.

Discussions (0)

[Suggest Correction](#)

Question 3 : 6406531245587

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Identify the correct scaling laws formula. The symbols have usual meaning

OPTIONS :

- ☐ $L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D}$
- ☐ $L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_D}{\alpha_N}} + \frac{D_c}{D} \right]^{\alpha_D}$
- ☐ $L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_N}$
- ☐ $L(N, D) = \left[\left(\frac{D_c}{D} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{N_c}{N} \right]^{\alpha_D}$
- ☐ None of these

Discussions (0)

[Suggest Correction](#)

[Broken Format?](#)

Question 4 : 6406531245588

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Suppose we are given a Model A with N layers and a dataset with D tokens. Choose the correct statements according to the scaling law.

OPTIONS :

- ☐ Increasing the model size requires a proportionate increase in the dataset size to reduce the test loss
- ☐ Increasing the model size requires a proportionate increase in the dataset size to reduce the training loss
- ☐ Increasing the model size requires us to double the dataset size to reduce the test loss
- ☐ Increasing the model size requires us to double the dataset size to reduce the train loss

Discussions (0)

[Suggest Correction](#)

[Broken Format?](#)

**Question 5 : 6406531245589**

View Solutions (0) Total Mark : 2.00 | Type : MCQ



The statement that KV caching is not helpful during inference in encoder models like BERT is

OPTIONS :

- ☐ TRUE
- ☐ FALSE

Discussions (0)

Suggest Correction

Broken Format?

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Question 6 : 6406531245590

View Solutions (0) Total Mark : 2.00 | Type : MCQ

The pre-attention score e_{ij} in Relative Position Encoding (RPE) can be written in matrix form as follows

$$E = XW_Q(XW_K + P_K)^T$$

What will be the dimension of P_K ? In all the options, T denotes context length.

OPTIONS :

- ☐ $T \times T \times T \times d_{model}$
- ☐ $T \times T \times d_{model}$
- ☐ $T \times d_{model}$
- ☐ $T \times T$
- ☐ None of these.

Discussions (0)

Suggest Correction

Question 7 : 6406531245591

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Consider the problem of length extrapolation using the Absolute Position Encoding (APE) scheme. Suppose the context length of a model during training is 512. Which of the following approaches allows the model to extrapolate to a context length of 1024 tokens

OPTIONS :

- ☐ Fixed Sinusoidal Encoding
- ☐ Parameterized APE
- ☐ ALiBi
- ☐ T5

Discussions (0)

Suggest Correction

Broken Format?

Question 8 : 6406531245592

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Select correct statements about sinusoidal positional encoding:



OPTIONS :

- ☐ It produces a unique encoding for each time-step (token's position in asentence)
- ☐ Distance between two tokens at time-steps 3 and 4 is independent of thelength of the input sequence.
- ☐ It is non-deterministic.
- ☐ None of these.

Discussions (0)

Suggest Correction

Broken Format?

Question 9 : 6406531245593

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Suppose we have a dataset for machine translation tasks with thousands of samples. Suppose a team considers training the transformer model. The model could be trained using two approaches

A. Autoregressive training B. Teacher forcing

Choose the correct statements

OPTIONS :

- ☐ Approach A helps the model to converge faster than approach B
- ☐ Approach B helps the model to converge faster than approach A
- ☐ One can start the training with approach B first and then switch to approach Aafter some training steps
- ☐ Once the training starts with approach A and then switching to approach Bafter some training steps cannot be done

Discussions (0)

Suggest Correction

Broken Format?

Question 10 : 6406531245595

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Select correct statements regarding character level tokenization:

Note: Assume that the character level vocabulary has English uppercase letters, lowercase letters, digits, special symbols, etc.

OPTIONS :

- ☐ The size of the vocabulary with character-level tokenization will be larger thanthat with word-level tokenization if the corpus consists of all the books published on Harry Potter (in English).
- ☐ The vocabulary will not expand if new English words/sentences are added tothe corpus.
- ☐ No issue of handling unknown tokens as long as the unknown tokens are fromEnglish vocabulary
- ☐ None of these.

Discussions (0)

Suggest Correction

Broken Format?

Question 11 : 6406531245597

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Consider the C4 pipeline. Which of the following will NOT pass through it as it is?

OPTIONS :

- ☐ ir a la escuela es un buen habito.
(Above sentence is Spanish translation of "Going to school is a good habit")

Following function computes $n - th$ Fibonacci number:

- ☐

```
def fibo(n):  
    if n <= 1:  
        return n
```



```
return fibo(n - 1)
+ fibo(n - 2)
```

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☐ D Gukesh defeated defending champion Ding Liren at the World

☐ Chess Championship. D Gukesh defeated defending champion
Ding Liren at the World Chess Championship.

☐ Sardar Vallabhbhai Patel was the first deputy prime minister
of independent India.

☐ None of these.

Discussions (0)

Suggest Correction

Broken Format?

Question 12 : 6406531245598

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following are design choices for building a large language model?

OPTIONS :

☐ Activation function in feed forward layer

☐ Training dataset.

☐ Positional encoding

☐ Attention mechanism.

☐ None of these.

Discussions (0)

Suggest Correction

Broken Format?

Question 13 : 6406531245594

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following set of GPT-based text generative models
(represented by M_i with $i \in \{1, 2, 3, 4\}$) and the corresponding text
generated with the prompt "I had to leave the party"

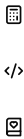
1. M_1 : I had to leave the party because had to leave the party.
2. M_2 : I had to leave the party leave leave
3. M_3 : I had to leave the party because because
4. M_4 : I had to leave the party because I received an SOS message
from my cousin

Select the model(s) that is(are) degenerative.

OPTIONS :

☐ M_1

☐ M_2



- ☐ M_2
- ☐ M_3
- ☐ M_4
- ☐ None of these.

Discussions (0)

[Suggest Correction](#)

Question 14 : 6406531245596

View Solutions (0) Total Mark : 3.00 | Type : MSQ

The knockout words in the passage given below denote the words to be dropped from the original sentence.

“These models can’t read ~~your mind~~. If outputs are too long, ~~ask for brief replies~~. If outputs are too simple, ask for expert-level writing. If you dislike the format, ~~demonstrate the format~~ you’d like to see. The ~~less the model has to guess~~ at what you want, the more likely you’ll get it.”

Which of the following represents the correct target sequence, with sentinel tokens, to the baseline model that uses pre-training denoising objectives? [z] represents the end of the sentinel token in a sentence. The characters inside the square brackets are the sentinel tokens.

OPTIONS :

- ☐ [a] your mind [b] ask for brief replies [c] demonstrate the format[d] less the model has to guess [z]
- ☐ [a] your mind [b] ask for brief replies [c] demonstrate the format[d] less the model has to guess
- ☐ [v] your mind [w] ask for brief replies [x] demonstrate the format[y] less the model has to guess
- ☐ [v] your mind [w] ask for brief replies [x] demonstrate the format[y] less the model has to guess [z]
- ☐ [a] your mind [b] demonstrate the format [c] ask for brief replies[d] less the model has to guess
- ☐ None of these

Discussions (0)

[Suggest Correction](#)

Question 15 : 6406531245599

View Solutions (0) Total Mark : 2.00 | Type : SA

How many entries of the unnormalized attention matrix QK^T will be calculated with the strided attention mechanism with $c = 5$. The sequence length is $T = 32$

Answer (Numeric):

Answer

Discussions (0)

[Suggest Correction](#)

Question 16 : 6406531245600

Total Mark : 0.00 | Type : COMPREHENSION

Consider following distribution generated by a GPT model after feeding the sentence “I love Manali because it is so”:

Token	Probability
beautiful	0.6
snowy	0.15
hilly	0.08
...	...



cold	0.05
good	0.01
bad	0.005
.	.
.	.

Consider following distribution generated by a GPT model after feeding the sentence "I love Manali because it is so":
Please note that the words are listed in decreasing order of their probabilities.
Based on the above data, answer the given subquestions

[Broken Format?](#)**Question 17 : 6406531245601**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If you are using the Top-P sampling method, what should be the minimum value of P to make sure that "hilly" could be selected as an output?

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 18 : 6406531245602**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

What value of P will allow "bad" to be selected in the output?

OPTIONS :

- ☐ 1
- ☐ 0.9
- ☐ 0.95
- ☐ 0.005
- ☐ 0.75
- ☐ 0.88

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 19 : 6406531245603**

Total Mark : 0.00 | Type : COMPREHENSION

Consider the following dictionary with the number of word occurrences in a corpus:

```
wo = { "taught" : 2,  
       "laughter" : 1,  
       "drought" : 4,  
       "tough" : 5}
```

Note: Append </w> to each word at the end.

Consider the following dictionary with the number of word occurrences in a corpus:

You will be working with word piece algorithm, answer the given subquestions in that context:

[Broken Format?](#)**Question 20 : 6406531245604**[View Parent QN](#)[View Solutions \(0\)](#)

**Question 20 : 6406531245604**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

How many tokens are there in the initial vocabulary?

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 21 : 6406531245605**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MCQ

Which of the following pairs has the least score before any merge?

OPTIONS :

☐ ('h', '</w>')☐ ('r', '</w>')☐ ('l', 'a')☐ ('r', 'o')☐ ('t', 'e')☐ ('u', 'g')[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 22 : 6406531245606**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

Which pair will be merged in the very first merge? Say the pair is ('a','b'), then enter **"ab"** (without quotes and white spaces). If there is a tie between two or more candidates, pick the one that occurs first in the original vocabulary outlined in the question.

NOTE: Enter the exact answer without any extra space in the beginning or at the end.

Answer (Alphanumeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 23 : 6406531245607**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

What is the score of the pair merged in the very first merge?

Answer (Numeric):

[Discussions \(0\)](#)[Suggest Correction](#)[Broken Format?](#)**Question 24 : 6406531245608**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MCQ

Assuming that the algorithm was stopped after two merges only, how will the word "later</w>" be



</>



tokenized with this vocabulary?

Please note that `</w>` is already appended to the word.

OPTIONS :

☐ la, t, er, `</w>`

☐ l, a, t, er, `</w>`

☐ la, t, e, r, `</w>`

☐ l, a, t, e, r, `</w>`

☐ l, a, ter, `</w>`

☐ lat, e, r, `</w>`

☐ None of these

Discussions (0)

[Suggest Correction](#)

Question 25 : 6406531245609

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions

The diagram below shows a few possible permutation blocks (for four attention heads) to be used in the masking matrix M in the block-wise attention mechanism.

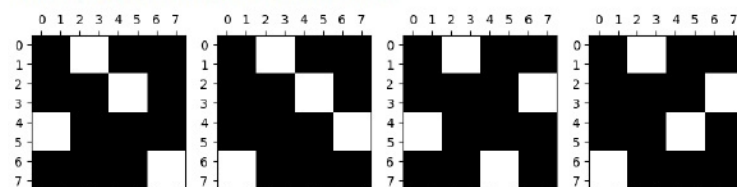


Figure 1: From left to right: a,b,c,d

[Broken Format?](#)

Question 26 : 6406531245610

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Of these 4 blocks, choose the block(s) that is (are) proper permutation of 4 blocks.

OPTIONS :

☐ a

☐ b

☐ c

☐ d

☐ None of these

Discussions (0)

[Suggest Correction](#)

[Broken Format?](#)

Question 27 : 6406531245611

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ



Let q_1, q_2, q_3, q_4 denote the 4 query blocks and k_1, k_2, k_3, k_4 denote the 4 key blocks. Select which of the following denotes the computation of pre-attention score for q_4 in the 4-th attention head

OPTIONS :

☐ $q_4^T k_4$

☐ $q_4^T k_1$

☐ $q_4^T k_2$

☐ $q_4^T k_3$

Discussions (0)

[Suggest Correction](#)

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